

Information loss of fixed health indicators as a control problem: simplex geometry and reachability limits

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Abstract

A health indicator compresses several condition-monitoring channels into one scalar, and normally keeps the same weights for the life of the component. Because degradation runs through several mechanisms at differing rates, the channel combination carrying most information about remaining life rotates as the component ages, so a fixed indicator necessarily discards information. We show that the operating point sets how much it discards, which makes that loss a control variable. The loss reduces to a single angle: the angle between the indicator and the direction along which information currently arrives. Writing each channel's share of that information as a probability vector places that angle on a statistical simplex. Zero loss is attainable exactly when the directions reachable at each instant share a common point, which for generic plants needs one fewer independently adjustable operating variable than there are mechanisms. Scaling every channel by the same factor leaves the loss unchanged, so no scalar summary of the Fisher information matrix can detect it. On a representative three-mechanism plant, steering the operating point cuts the discarded information by a factor of 68, and one dimensionless ratio, computable before the designer fixes any indicator, indicates whether a gain of that order is plausible. Across six public run-to-failure fleets the informative direction turns 30.7° over the life of a lithium-ion cell fleet and, against a negative control, 439 simulated turbofan records confirm that the operating point moves it. The same fleets bound the theory:

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the highest-fidelity turbofan data reject the separable plant form under which the operator’s authority becomes computable offline, so that branch describes a model class rather than measured machinery.

Keywords: prognostics, health indicator, remaining useful life, information loss, optimal control, reachability

1. Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record; the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [4, 2]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

The rotation is not a property of the instrumentation. The degradation physics produces it, and that physics responds to stress: temperature, load and duty cycle change the relative rates of the competing mechanisms, not only the overall speed. Running a cell hotter does not only age it faster; it also shifts the balance between the mechanisms, and therefore moves the direction in which information about remaining life arrives.

The operating point is, in most applications, something the operator chooses. So the misalignment between a fixed indicator and the informa-

31 tive direction is not fixed by nature. It is a function of the input, and the
32 operator can steer it.

33 That reframing turns a data-analysis question into a control question, and
34 control questions come with sharper answers than design heuristics do. One
35 can ask what misalignments are reachable at a given operating envelope,
36 whether zero loss is attainable at all, how many independent inputs that
37 would take, and what the best achievable loss is when it is not. These are
38 the questions this paper answers.

39 They sit against five bodies of work. Four address parts of the problem
40 and the fifth supplies the geometry used in Section 3; to our knowledge none
41 of them addresses the position taken here.

42 The prognostics literature treats the compression as a design problem to
43 be solved from data: searching genetically over candidate features against
44 monotonicity and trendability criteria [2], learning it with a recurrent net-
45 work [5], or surveying the resulting pipelines [1, 6]. Two limits on that
46 programme are known, information-theoretic bounds on what any fixed com-
47 pression can retain [3] and the number of run-to-failure units needed to learn
48 one reliably [4]. The first is our own, and the boundary is worth drawing
49 explicitly: it takes one degradation trajectory as given and asks how much
50 information a fixed compression must lose along it. The question here is the
51 complementary one, whether that trajectory can itself be steered so that the
52 same fixed map loses less. Both prior limits hold the trajectory fixed; neither
53 asks whether it can be changed.

54 The predictors downstream are overwhelmingly stochastic — Wiener,
55 gamma and inverse-Gaussian degradation processes with first-passage fail-
56 ure [7, 8], reviewed for remaining-life estimation in [9, 10] and embedded in
57 system-level design methodology in [11]. That literature models how damage
58 evolves, not how the measurement should be compressed before the model is
59 applied, which is the question upstream of it.

60 A separate literature does treat the operating point as a decision variable,
61 but to extend remaining life rather than to preserve estimability [12, 13];
62 the two aims can conflict, and Section 6 exhibits an instance in which they
63 disagree. Choosing inputs to make a state easier to estimate is itself classical,
64 going back to dual control [14], and its modern forms shape a scalarisation
65 of the Fisher information matrix [15, 16, 17]. Our Proposition 4.6 shows the
66 quantity at issue here to be invisible to every such scalarisation.

67 The fifth body is information geometry. The reformulation of Section 3
68 rests on the Fisher–Rao geometry of the probability simplex and on the

Bhattacharyya coefficient [18, 19], and the flow it produces is the replicator equation of evolutionary dynamics [20, 21]. These are standard objects; what is new here is their identification with the informative direction of a degrading component, and the fact that the fitness driving the flow is actuated.

The contributions are as follows.

1. The information a fixed indicator discards is an exact function of one misalignment angle (Lemma 2.1). Placing the channels' shares of the Fisher information on a statistical simplex turns the degradation into a replicator flow whose rates the operating point modulates, and the optimal indicator into a barycentre of the degradation path (Section 3).
2. Zero loss is attainable exactly when the reachable direction sets share a point, which generically needs at least as many independent operating variables as mechanisms less one (Section 4). Supplying the missing input collapses the peak reachability floor from 4.52° to zero at every state, so the rule prices instrumentation rather than only bounding it. Taking logarithms makes every reachable set a translate of one fixed set, which turns the question into a containment and shows the counting rule to be sufficient but not necessary; the exceptions exist for any number of mechanisms and are too fragile to build on (Section 4.1). The count separates two objectives that are easily conflated: holding the misalignment at a constant level is a single scalar condition and generically needs one input, whereas freezing the informative direction outright is $d - 1$ conditions. That gap is the difference between useful steering and lossless steering, and an installation with one controllable variable can have the first without the second.
3. The loss is invariant to a common gain on all channels and is therefore invisible to every scalarisation of the Fisher information matrix, so observability-driven criteria cannot be used to choose an indicator.
4. The best achievable loss is the value of an optimal control problem whose objective is a ratio, and whose terminal time is itself controlled (Section 5), and a dimensionless regime index, computable before any indicator is designed, screens for the regime in which steering can have substantial value (Section 6.7).
5. The premises the construction rests on are separately testable on public run-to-failure data, and Section 7 tests them in three stages: whether the informative direction rotates at all, whether the operating point moves it, and whether the separable stress model survives. Two survive

106 and one does not, which makes that section a test of which hypothe-
 107 ses withstand contact with data rather than a demonstration that the
 108 method works. The rotation is present in measured fleets; the actua-
 109 tion premise is so far confirmed on simulation only; and higher-fidelity
 110 turbofan data reject the separability of Lemma 4.3, which places that
 111 lemma, and the offline route it opens, as statements about a model
 112 class rather than about measured machinery.

113 Section 2 states the model. Sections 3 to 5 develop the theory, Section 6
 114 exercises the three of its claims that are stated sharply enough to be refuted,
 115 and Section 7 reports what public fleet data can and cannot establish about
 116 the premises the theory rests on.

117 2. Problem formulation

118 This section states the degradation model, says precisely what a prognos-
 119 tic algorithm is being asked to estimate and where the uncertainty in that
 120 quantity comes from, derives the direction along which information about
 121 it arrives, and defines the fixed indicator whose information loss the rest of
 122 the paper studies. Readers familiar with [3] will recognise the model; the
 123 derivation is given in full here because the geometry of Section 3 depends on
 124 several of its details.

125 2.1. Degradation driven by a controlled operating point

126 A component degrades through d distinct mechanisms acting at once. In
 127 a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-
 128 electrolyte interphase and mechanical fatigue of the electrode; in a rolling
 129 bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the
 130 damage accumulated by mechanism i up to time t , and collect these into the
 131 state $x = (x_1, \dots, x_d)$.

132 One externally chosen operating point drives all d mechanisms. It collects
 133 the temperature, the mechanical or electrical load, and the duty cycle at
 134 which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

135 with $\mathcal{U}$ compact, representing the operating envelope the application permits.
 136 Throughout this paper u is the control input: it is the decision variable,
 137 and the question we ask is what choosing it well can buy. The number of

138 independent inputs, m , turns out to decide what is achievable, and it is
 139 usually much smaller than the number of mechanisms d .
 140 The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

141 in which the two factors play different roles:

- 142 • $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much
 143 faster that mechanism runs at operating point u than at a reference
 144 point. It carries all the dependence on the input.
- 145 • $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends
 146 on the damage it has already done. It carries all the dependence on
 147 the state.

148 This separation is the standard accelerated-life form and covers the stress
 149 models in routine use [22]. In the Arrhenius model the input is absolute
 150 temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential
 151 constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's
 152 constant. In the inverse-power-law model the input is a mechanical or electri-
 153 cal stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress
 154 exponent. Choosing h_i recovers the exponential, power-law and stretched-
 155 exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and
 156 $h_i > 0$.

157 The within-unit dynamics are deterministic, where most of the degrada-
 158 tion literature places the randomness inside a single unit's path [7, 8, 10].
 159 We put it between units instead (Section 2.2). The object of study is the
 160 direction along which information arrives, which is fixed by the relative rates
 161 of the competing mechanisms: a within-unit diffusion would jitter that di-
 162 rection without altering its systematic motion, whereas dispersion in the rate
 163 parameters across a population is what makes remaining life uncertain and
 164 therefore worth estimating.

165 We measure damage on a scale normalised so that mechanism i exhausts
 166 the component's tolerance when x_i reaches unity. A component is withdrawn
 167 at a condemnation limit at or below that point, so we write the end of life
 168 as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

169 No result below depends on the value of x_f , which enters only by fixing where
 170 a trajectory stops. The computed quantities of Section 6 do depend on it,
 171 and the value used there is stated with the rest of the test plant.

172 2.2. What is to be estimated, and why it is uncertain

173 The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

174 the time left before failure.

175 We must say where the uncertainty in R lies, because (2) is a deterministic
 176 differential equation: for one component whose initial damage and paramet-
 177 ers were known exactly, T_f and hence R would be known exactly, and there
 178 would be nothing to estimate. The uncertainty is between components, not
 179 within one. Units built to the same specification differ in initial damage and
 180 in the coefficients of (2), which is the unit-to-unit variability that reliability
 181 engineering treats as the primary source of scatter in life data [22]. A prog-
 182 nostic algorithm observes one unit drawn from that population and must
 183 infer its remaining life.

184 We therefore treat R , at any given observation instant, as a random
 185 variable over the population of units, with prior variance $\text{Var}[R] > 0$. This
 186 variance is the uncertainty the algorithm exists to reduce, and it is the reason
 187 the estimation problem is not trivial. Everything below is written for a
 188 nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

189 2.3. Observations and the informative direction

190 Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

191 with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal
 192 variables. The record available to the algorithm is the collection of these
 193 measurements over the observation window.

194 Following [3], we linearise the dependence of the record on R about the
 195 nominal trajectory, which makes the record and R jointly Gaussian. The
 196 sensitivity of channel i to remaining life follows from (4): because R falls at
 197 unit rate as t advances,

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

Dividing each channel by its own noise level puts them on a common scale,
and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = \left[\sigma_i^{-1} \varphi_i(u) h_i(x_i) \right]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d} / \|\mathbf{d}\|. \quad (7)$$

In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the *informative direction*, the combination a health indicator has to be aligned with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much information arrives and $\hat{\mathbf{d}}$ is where it arrives from, and the loss of a fixed indicator depends only on the second. Experiment design in the observability tradition scores the first, which is why Proposition 4.6 finds it blind to the loss.

We use two properties of (7) repeatedly below.

Positivity. Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i > 0$ in (2). This is a consequence of the degradation model rather than an assumption imposed for convenience, and Remark 3.1 shows that the results below would fail without it.

Actuation. $\mathbf{d}$ is an explicit algebraic function of the state and the input. The direction along which information about R arrives is therefore not a fixed property of the instrumentation: it moves when the operating point moves. That is the observation this paper is built on, and Fig. 1 places it: the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.

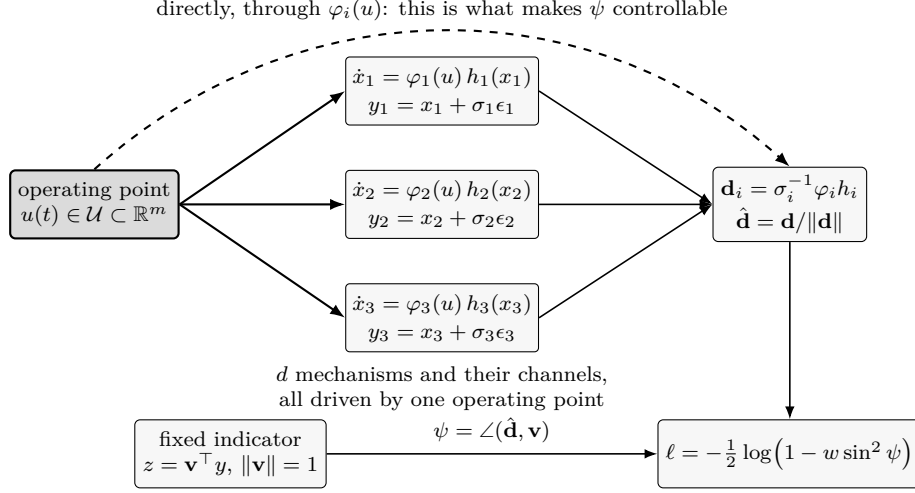
Because R is scalar, a single signal-to-noise ratio governs the information the record carries about it, namely the squared sensitivity times the prior uncertainty,

$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

Rates in (7) are taken with respect to clock time, whereas [3] differentiates with respect to age normalised by T_f . The two conventions differ by a factor common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same geometry in Sections 3–4, while rescaling γ . Section 5, where lifetime enters the objective, restores the normalised convention.

2.4. The fixed health indicator and its information loss

A health indicator compresses the d measured channels into one scalar before any life prediction is made. We take it to be a fixed linear map.



Alignment asks $\hat{\mathbf{d}}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

230 Since (7) already measures each channel in units of its own noise, we write
 231 the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and
 232 $\tilde{y} := \Sigma^{-1/2}y$ the whitened record,

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$

233 with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole
 234 life of the component. Fixing it is not a modelling convenience: the map
 235 is typically embedded in firmware or frozen by certification, so it cannot be
 236 re-tuned as the component ages. An indicator specified on the raw measure-
 237 ments, $z = \mathbf{w}^\top y$, is the same map under $\mathbf{v} \propto \Sigma^{1/2}\mathbf{w}$, and the normalisation
 238 $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters
 239 because the angle below is measured after whitening: unless the σ_i are equal,
 240 a unit vector in raw coordinates is not a unit vector in whitened ones, and
 241 the two readings of ψ disagree.

242 We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle
 243 between the informative direction $\hat{\mathbf{d}}(t)$ and the indicator weights $\mathbf{v}$. Af-
 244 ter whitening, the signal in the record points along $\hat{\mathbf{d}}$ and has magnitude

245 $\|\mathbf{d}\|\sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit
 246 vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise,
 247 so the indicator's signal-to-noise ratio is

$$\gamma \cos^2 \psi \quad (10)$$

248 against the γ of (8) available to an oracle that is free to re-aim its projection at
 249 $\hat{\mathbf{d}}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio
 250 s carries $\frac{1}{2} \log(1 + s)$ nats about R [23], the information the fixed indicator
 251 forfeits, at each instant, is

$$\ell = \frac{1}{2} \left[\log(1 + \gamma) - \log(1 + \gamma \cos^2 \psi) \right] \geq 0, \quad (11)$$

252 The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

253 a loss per unit time. We optimise this rate rather than the information the
 254 whole observation record forfeits, because (12) is unaffected by how densely
 255 the record is sampled and is therefore a property of the indicator rather
 256 than of the measurement schedule. The two are different quantities, and
 257 Proposition 2.2 gives the second in closed form.

258 Two remarks on what (11) does and does not say. It is a loss relative to
 259 an oracle that no fixed map can realise, so it measures the price of the time-
 260 invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still
 261 pays it whenever $\hat{\mathbf{d}}$ moves. And it depends on the indicator only through the
 262 single angle ψ : two indicators that make the same angle with $\hat{\mathbf{d}}$ are equally
 263 good, whatever else distinguishes them.

264 The next lemma replaces (11) by a closed form that will be more conven-
 265 nient throughout, and records how it behaves at small misalignment.

266 **Lemma 2.1.** *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

267 *exactly and at every signal-to-noise ratio, and*

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

268 *Proof.* Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log (1 - w \sin^2 \psi).$$

269 For (14), use $-\frac{1}{2} \log(1-s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 -$
 270 $\psi^4/3 + O(\psi^6)$; collecting the ψ^4 terms gives $-w/6 + w^2/4 = w(3w-2)/12$. $\square$

271 The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is
 272 therefore the leading term of an exact identity rather than an approximation
 273 of unknown quality. The quartic coefficient $w(3w-2)/12$ changes sign at
 274 $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic
 275 model overstates the loss, above it the model understates it. This matters in
 276 practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life
 277 measurement with a weak signal on one side, a well-instrumented component
 278 near failure on the other. Fig. 2(a) plots the exact loss against its quadratic
 279 approximation on both sides of that ratio.

280 **Proposition 2.2.** *Let the record consist of measurements (5) at instants*
 281 *$t_1, \dots, t_N$, with the noise independent across channels and across instants,*
 282 *and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1+G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

283 *Then the information about R that the fixed indicator forfeits over the whole*
 284 *record is*

$$\ell_{\text{rec}} = -\frac{1}{2} \log (1 - WS). \quad (15)$$

285 *Proof.* In whitened coordinates the linearised record is $\tilde{y}_k = \mathbf{d}_k R + \epsilon_k$ with
 286 $\epsilon_k \sim N(0, I_d)$ independent, so the Fisher information it carries about R is
 287 $\sum_k \mathbf{d}_k^\top \mathbf{d}_k = \sum_k \|\mathbf{d}_k\|^2$ and an oracle reading the whole record obtains $\frac{1}{2} \log(1 +$
 288 $G)$ nats. The fixed indicator returns the scalars $z_k = \mathbf{v}^\top \tilde{y}_k = (\mathbf{v}^\top \mathbf{d}_k) R + \mathbf{v}^\top \epsilon_k$,
 289 whose noise terms $\mathbf{v}^\top \epsilon_k$ are standard normal and independent because $\|\mathbf{v}\| =$
 290 1 , so the compressed record carries Fisher information $\sum_k (\mathbf{v}^\top \mathbf{d}_k)^2$. Since
 291 $\mathbf{v}^\top \mathbf{d}_k = \|\mathbf{d}_k\| \cos \psi_k$,

$$\text{Var}[R] \sum_k (\mathbf{v}^\top \mathbf{d}_k)^2 = \text{Var}[R] \sum_k \|\mathbf{d}_k\|^2 \cos^2 \psi_k = G(1 - S),$$

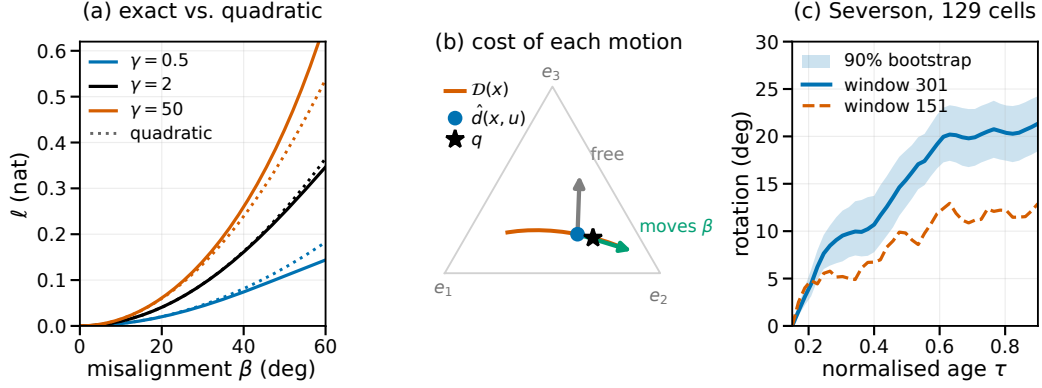


Figure 2: (a) Lemma 2.1: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

and subtracting the two mutual informations gives $\frac{1}{2} \log(1 + G) - \frac{1}{2} \log(1 + G(1 - S)) = -\frac{1}{2} \log \frac{1+G-GS}{1+G} = -\frac{1}{2} \log(1 - WS)$. $\square$

Equation (15) is (13) again, with the instantaneous signal-to-noise ratio γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore survives the passage from an instant to a record. What does not survive is the averaging: $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ. They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises $\mathbf{v}^\top (\sum_k \mathbf{d}_k \mathbf{d}_k^\top) \mathbf{v}$, so the best fixed indicator for the record is the leading eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$ and should not be read as a statement about (15).

3. The information simplex and its replicator flow

Lemma 2.1 reduced the loss to one angle. This section shows that the angle lives on a familiar object, the probability simplex, and that the degradation traces a flow along it that the operating point drives. The payoff is threefold: the loss becomes a recognised statistical distance, the optimal indicator becomes a barycentre, and the effect of the input becomes a single interpretable quantity. Two of the results here are load-bearing for what

311 follows — the replicator form of Theorem 3.4 and the misalignment rate of
 312 Theorem 3.6, which Section 4 builds the reachability argument on. The Bhat-
 313 tacharyya identity and the barycentre are interpretation: they say what the
 314 loss and the optimal indicator already are in recognised terms, and nothing
 315 later depends on them.

316 3.1. Information shares

317 Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components
 318 of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

319 The interpretation is direct. The total Fisher information the record carries
 320 about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i
 321 is exactly the share of the total information about remaining life that arrives
 322 through channel i . A component in which one mechanism dominates the
 323 degradation has p near a vertex of the simplex; one in which three mecha-
 324 nisms contribute equally sits near its centre.

325 The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an
 326 optimal indicator may be taken with non-negative weights.

327 **Remark 3.1** (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$
 328 at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative
 329 weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$
 330 inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-
 331 indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a
 332 third of the time. The results therefore require the positive-orthant structure
 333 of the degradation model; it is not only a notational convenience.*

334 3.2. The loss as a Bhattacharyya distance

335 The identification below is used twice in what follows: it gives “the aver-
 336 age of a curve on the simplex” an unambiguous meaning (Section 3.3), and
 337 it lets the high-signal limit of the loss be read off as a known divergence.

338 **Proposition 3.2.** $\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and
 339 q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on
 340 the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya
 341 divergence,

$$\ell = -\frac{1}{2} \log \left[(1 - w) + w e^{-2D_{\text{B}}(p, q)} \right]. \quad (17)$$

342 As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.

343 *Proof.* By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

344 which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$
 345 is the classical isometry between the simplex under the Fisher–Rao metric
 346 and the positive orthant of the unit sphere, under which the geodesic distance
 347 is twice the spherical angle [18, 19]; hence $\psi = \frac{1}{2}d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$,
 348 and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

349 The square root is a bijection between the simplex and the non-negative
 350 part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition
 351 it is an isometry. We use the two descriptions interchangeably from here
 352 on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable
 353 direction, and a distance between points of the simplex is the angle between
 354 the corresponding unit vectors.

355 The resulting picture is the one the rest of the paper develops. The
 356 degrading component is a curve on the simplex, the health indicator is a
 357 single point on it, and the information lost depends only on the distance
 358 between them. Design of the indicator is the choice of one point; control of
 359 the operating point is the shaping of the curve.

360 3.3. The optimal indicator as a barycentre

361 If the indicator is one point and the plant is a curve, the best point should
 362 be some kind of average of the curve. It is, but of two different kinds in two
 363 different limits.

364 **Proposition 3.3.** *Let q^* minimise the time-averaged loss.*

- 365 (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}$ pointwise, so q^* tends to the Bhattacharyya*
 366 *barycentre $\arg \min_q \int D_{\text{B}}(p_\tau, q) d\tau$.*
- 367 (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so q^* tends to the w -weighted*
 368 *Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}\bar{w} \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} :=$*
 369 *$\min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

370 *Proof.* (i) As $\gamma \rightarrow \infty$ we have $w \rightarrow 1$, and (13) gives

$$\ell \rightarrow -\frac{1}{2} \log \cos^2 \psi = -\log \sum_i \sqrt{p_i q_i} = D_{\text{B}}$$

by Proposition 3.2. The convergence is uniform, which is what carries the minimisers with it. Positivity keeps p_τ interior, and $\sqrt{q_i} \geq q_i$ with $\sum_i q_i = 1$ gives $\cos \psi = \sum_i \sqrt{p_i q_i} \geq \min_i \sqrt{p_i}$ for every q in the simplex, so $\cos \psi$ is bounded away from zero uniformly in q and in τ and D_B stays finite. Both objectives are continuous on a compact set and uniformly close on it, so their minimisers converge. (ii) By (14), $\bar{\ell} = \frac{1}{2} \bar{w} \mathbb{E}_\mu[\psi^2] + O(\psi^4)$ with $d\mu = w d\tau / \bar{w}$; minimising a μ -weighted mean square of a geodesic distance is the definition of the weighted Fréchet mean, and $\psi = \frac{1}{2} d_{\text{FR}}$ converts $\frac{1}{2} \bar{w} \mathbb{E}_\mu[\psi^2]$ into $\frac{1}{8} \bar{w} \mathbb{E}_\mu[d_{\text{FR}}^2]$. $\square$

The two limits are taken in different parameters, signal-to-noise ratio in (i) and arc length in (ii), and they do not coincide. On a curve of angular span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5} degrees of the latter by 50 dB. For design purposes the practical consequence is mild: either barycentre of a nominal degradation path is a good starting indicator.

3.4. The replicator flow and the role of the input

We now ask how fast the informative direction moves and what the operating point does to that motion.

Theorem 3.4. *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i . Along (2),*

$$\dot{\mathbf{d}}_i = \hat{\mathbf{d}}_i (\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

so the information shares obey the replicator equation of evolutionary dynamics, in which each share grows in proportion to the amount by which its own rate exceeds the average, the average being taken over the shares themselves:

$$\dot{p}_i = 2p_i (\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

and

$$\|\dot{\mathbf{d}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = \left[\varphi_i(u) h'_i(x_i) \right]_i + \nabla_u \log \varphi \dot{u}. \quad (20)$$

Proof. From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1} (\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has components $\hat{\mathbf{d}}_i \dot{g}_i - \hat{\mathbf{d}}_i \sum_j \hat{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives $\text{Var}_p[\dot{g}]$. Since $p_i = \hat{\mathbf{d}}_i^2$, $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\mathbf{d}}_i = 2p_i (\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i) \varphi_i h_i$, and $(h'_i/h_i) \varphi_i h_i = \varphi_i h'_i$. $\square$

Equation (20) has a compact reading. The speed at which the informative direction rotates equals the information-weighted spread of the mechanisms' logarithmic rates. If all mechanisms accelerate at the same proportional rate the direction does not move at all, however fast the component is degrading; it is disagreement among the mechanisms that rotates it. The identity is the Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's fundamental theorem of natural selection [18, 20, 21]. What is new here is its identification with the informative direction of a degrading component, and, decisively, that the quantity playing the role of fitness, the logarithmic rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's lever.

Remark 3.5 (contrast with Oja's flow). *Oja's flow for online principal-component extraction has the same algebraic form; for a diagonal covariance it reduces to a replicator equation with fitness twice the eigenvalues [24]. The roles are opposite. There the flow is an algorithm, run to fixation in order to find a direction, with constant fitness. Here it is the plant, its fitness is time-varying and actuated, and the aim is to prevent fixation from carrying the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\hat{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.1 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{e}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\hat{\mathbf{d}}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$. Also $\hat{e} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}})/\sin \beta$ and $\hat{\mathbf{d}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}/\sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\hat{\mathbf{d}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits

below hold for every measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits.

The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d - 2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

4. Reachability limits on the misalignment

Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The distinction is the subject of this section, and it is what decides whether a fixed indicator can be made lossless at all.

For a state x , the set of informative directions the operator can produce by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

the reachable direction set, and the closest the indicator can be brought at that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures the actuation authority available at x , and it shrinks to a point when the mechanisms respond identically to the input.

Theorem 4.1. *Fix an admissible input and let $x(\cdot)$ be the trajectory it generates. Zero loss along that trajectory is attainable with the fixed indicator q if and only if $q \in \cap_t \mathcal{D}(x_t)$. Solving $\hat{\mathbf{d}}(x, u) = q$ for u imposes $d - 1$ independent scalar equations on m unknowns, so for generic data a common point exists only if $m \geq d - 1$. If $m < d - 1$, then for generic data $\beta_{\min} > 0$ along the trajectory; Remark 4.5 shows the qualifier cannot be dropped. In every case $\bar{\ell} \geq \int \ell(\beta_{\min}(x_t, q)) dt / T_f$ for that trajectory.*

Proof. By Lemma 2.1, $\ell(\beta) = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\beta_t = 0$ for almost every t , that is iff $\hat{\mathbf{d}}(x_t, u_t) = q$, that is iff $q \in \mathcal{D}(x_t)$ for every t . Solving $\hat{\mathbf{d}}(x, u) = q$ imposes $d - 1$ independent scalar equations, both sides being unit vectors, on m unknowns. If $m < d - 1$ then $\mathcal{D}(x)$ is the image of an

463 m -dimensional set, hence of dimension at most m , and is a null subset of the
 464 $(d - 1)$ -dimensional sphere that a generic q misses; if $m \geq d - 1$ the implicit
 465 function theorem supplies a solution wherever the Jacobian has full rank.
 466 The inequality $\beta_t \geq \beta_{\min}(x_t, q)$ holds for every admissible input by definition
 467 of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and is strict exactly where $q \notin \mathcal{D}(x_t)$, which
 468 for $m < d - 1$ is generic. Since ℓ is increasing on $[0, \pi/2]$, the bound follows
 469 on dividing by T_f . $\square$

470 This is a state-conditioned result. The bound is genuine, in the sense
 471 that no input violates a reachability constraint at the state it occupies, but
 472 it is conditional on the trajectory, which the input also steers. Optimising
 473 over the trajectory as well is deferred to Section 5, and the two levels are the
 474 structure of the argument: a state-conditioned limit here, a trajectory-level
 475 limit there.

476 The practical reading of Theorem 4.1 is a counting rule: a component
 477 degrading through d mechanisms generically needs $d - 1$ independently ad-
 478 justable operating variables before a fixed indicator can be kept perfectly
 479 aligned. Section 4.1 says what the qualifier covers. Three mechanisms and a
 480 single temperature setpoint will not do it. But a weaker and much cheaper
 481 objective is available.

482 **Corollary 4.2** (what actuation is actually required). *Two objectives that are*
 483 *easily conflated carry different requirements.*

- 484 (i) *Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d - 1$*
 485 *conditions on $\dot{u}$, hence generically $m \geq d - 1$.*
- 486 (ii) *Holding β at a constant level, not necessarily zero, is the single scalar*
 487 *equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$*
 488 *stays in $\mathcal{U}$; generically $m \geq 1$ suffices.*

489 *Proof.* (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for every t , so $\hat{\mathbf{d}}$ is constant;
 490 by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d - 1$ independent
 491 conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap
 492 between them is the content of that theorem: $\dot{\beta}$ sees only the component of
 493 $\dot{\hat{\mathbf{d}}}$ along $\hat{e}$, so the remaining $d - 2$ directions of rotation are unconstrained by
 494 (ii) and constrained by (i). $\square$

495 Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the
 496 trajectory that carries it, and is the problem of Section 5.

497 4.1. The reachable sets are translates of one set

498 Theorem 4.1 counts equations against unknowns and hedges with
 499 “generic”. The hedge is not idle, and the following observation says exactly
 500 what it covers.

501 **Lemma 4.3** (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo*
 502 *the all-ones vector, which is what $\hat{\mathbf{d}}$ retains. Then*

$$\begin{aligned}\xi(x, u) &= a(x) + b(u), \\ a_i(x) &= \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u),\end{aligned}\tag{23}$$

503 *both taken modulo the all-ones vector. Consequently every reachable direction*
 504 *set is the same set translated: $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.*

505 *Proof.* Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the
 506 product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along
 507 the all-ones vector. $\square$

508 The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the
 509 stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator’s authority
 510 can therefore be computed offline from the accelerated-life model, with no
 511 degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage
 512 shapes h and the noise levels σ , and every reachable set is that one set carried
 513 along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the
 514 plant’s demand $\mathcal{A}$ sticks out of the operator’s authority $\mathcal{B}$.

515 That offline route is the most useful consequence drawn in this paper
 516 and also the most restrictive. It holds on the separable subclass and not
 517 beyond it, and the best high-fidelity turbofan data we have reject separability
 518 (Section 7). This subsection and Proposition 4.4 should therefore be read as
 519 describing a structured subclass that is convenient when it applies, not as
 520 a general account of real machinery. Nothing stated before this subsection
 521 depends on the assumption – Theorem 4.1 included – and Fig. 6 marks the
 522 one branch of the development that does.

523 **Proposition 4.4** (exact criterion). *Zero loss is attainable along a trajectory*
 524 *if and only if $\mathcal{A}$ is contained in a translate of $-\mathcal{B}$. In particular $\mathcal{A}$ must lie*
 525 *in an affine subspace of dimension at most $\dim \mathcal{B} \leq m$, which for a curve*
 526 *in the $(d - 1)$ -dimensional quotient fails generically unless $m \geq d - 1$; this*
 527 *recovers Theorem 4.1. When it does not fail, $m < d - 1$ suffices.*

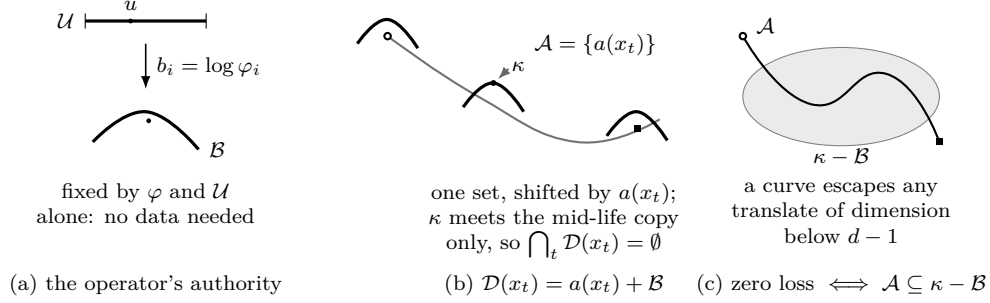


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the whole curve to lie in one translate (Proposition 4.4).

528 *Proof.* By Theorem 4.1, zero loss holds iff $q \in \mathcal{D}(x_t)$ for every t . Writing κ
529 for the quotient class of $\log \mathbf{v}$ and using Lemma 4.3, that reads $\kappa - a(x_t) \in \mathcal{B}$
530 for every t , that is $\mathcal{A} \subseteq \kappa - \mathcal{B}$. The dimension statement follows because a
531 translate of $\mathcal{B}$ spans an affine subspace of dimension $\dim \mathcal{B}$. $\square$

532 The containment is drawn in Fig. 3(c), and the mechanism the count
533 obscures is now visible. Alignment at an instant fixes u_t outright, since
534 $b(u_t) = \kappa - a(x_t)$ determines it; the input is spent, and any residual constraint
535 on the state must then hold along the flow with no freedom left to enforce it.
536 One input can carry the whole burden only when no residual constraint
537 is left, which happens when $\mathcal{A}$ already lies parallel to $\mathcal{B}$.

538 **Remark 4.5** (a family with $m = 1$ and any d). Take exponential shapes
539 $h_i(x) = e^{\nu_i(x_i - x_i^0)}$, which (2) admits, and Arrhenius factors. Under the ramp
540 $u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 =$
541 $\varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = rE_i/\varphi_i^0$ makes
542 that velocity parallel to E , which is the direction of $\mathcal{B}$, and $b(u_t)$ retreats
543 along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Integrating the
544 construction for $d = 2$ to 8 with one input holds the misalignment below
545 1.5×10^{-6} degrees and the mean loss below 10^{-15} . That is against 1.2×10^{-5}
546 to 5.4×10^{-4} for the best constant input on the same plant.

547 This is a statement about the sharpness of Theorem 4.1 and not a way
548 to save an actuator. The family of Remark 4.5 is a knife edge. Perturbing
549 the shape coefficients ν by 1% raises the mean loss to 3.2×10^{-4} under the
550 myopic policy, against 1.8×10^{-4} for the best constant input on the same

551 perturbed plant. At one percent of shape error, steering is already worse than
 552 not steering. Closing the loop halves the damage but does not repair it, so
 553 the fragility belongs to the construction and not to the open-loop schedule.
 554 Mistuning the activation energies instead is tolerated to about two percent. A
 555 designer should read Proposition 4.4 as an explanation of when the counting
 556 rule binds, and Theorem 4.1 as the rule to design against.

557 4.2. Invisibility to Fisher-information criteria

558 A natural objection is that all of this should follow from designing the in-
 559 put to maximise the Fisher information about R , as in the observability-aware
 560 literature discussed in Section 1. It does not, and the reason is structural.

561 **Proposition 4.6** (Fisher blindness). *The loss of a fixed indicator is not a*
 562 *function of the Fisher information about R : configurations with equal Fisher*
 563 *information can have different losses, and configurations with equal losses*
 564 *can have different Fisher information. No criterion depending on the Fisher*
 565 *information alone can therefore rank fixed indicators by the loss they incur.*

566 *Proof.* For a scalar latent the Fisher information is the number $\|\mathbf{d}\|^2 \text{Var}[R]$,
 567 so trace, determinant and least eigenvalue coincide and every scalarisation is
 568 a strictly monotone function of it. Under the rescaling $\mathbf{d} \mapsto c\mathbf{d}$ with $c > 0$,
 569 that number changes while $\hat{\mathbf{d}}$, and hence β and ℓ , do not. A common gain
 570 therefore equalises the scores of two configurations with different losses, and
 571 separates the scores of two configurations with the same loss. $\square$

572 Stated in engineering terms, a Fisher criterion measures the strength of
 573 the signal, whereas the loss of a fixed indicator depends on its direction.
 574 Amplifying every channel by the same factor increases the first and leaves
 575 the second unchanged. Section 6 exhibits this concretely: with the Fisher
 576 score held equal to seven digits, the loss still spans a factor of 9.4.

577 5. The best achievable loss as a fractional control problem

578 Theorem 4.1 bounds the loss along a given trajectory. But the input
 579 steers the trajectory as well as the misalignment on it, so a fundamental
 580 limit requires optimising over both at once. This changes the structure of
 581 the problem.

582 The objective is a time average over a lifetime that the input itself deter-
 583 mines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (24)$$

584 Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes
 585 (24) a fractional optimal control problem, meaning that the objective is a
 586 ratio rather than that the dynamics carry a fractional-order derivative. The
 587 fraction is not a technicality: it is the source of the trade-off, as follows.

588 Suppose T_f were fixed. Nothing in (2) penalises the input directly, so
 589 the answer would be trivial: find whichever admissible u minimises β and
 590 stay there. Because T_f is itself controlled, the denominator of (24) rules that
 591 solution out. The operating point that aligns the indicator is in general not
 592 the operating point that prolongs life, so aligning now shortens the interval
 593 over which the alignment is averaged. The optimum must trade the two,
 594 and Section 6.5 measures how steep that trade is on a concrete plant. A
 595 fixed-horizon formulation represents none of it, which is why one was tried
 596 and discarded (Section 6.5).

597 5.1. A converse bound uniform over trajectories

598 Before computing anything we record a certificate that removes the con-
 599 ditioning. Theorem 4.1 bounds the loss along a trajectory the input has
 600 already chosen; the following proposition bounds it over all of them at once,
 601 and it is what makes the number computed below a limit rather than an
 602 observation.

603 **Proposition 5.1** (verification certificate). *Fix an indicator q and a constant*
 604 *$\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood*
 605 *of the admissible state region such that*

- 606 (i) $V(x) \leq 0$ for every x on the failure surface $\{\max_i x_i = 1\}$, and
- 607 (ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible
 608 state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (25)$$

609 Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if
 610 $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence
 611 over all trajectories.

612 *Proof.* Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1
613 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and
614 the chain rule gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost
615 every t . Applying (25) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e.,
616 and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

617 the last step by (i), since x_{T_f} lies on the failure surface. Dividing by $T_f > 0$
618 gives the claim. $\square$

619 Condition (25) constrains the plant data alone: it quantifies over every
620 state and every admissible input, and mentions no trajectory. Any V one
621 can exhibit therefore certifies a limit, and the search for a converse becomes
622 the search for a verification function.

623 The C^1 hypothesis cannot be weakened to Lipschitz continuity. A Lip-
624 schitz V is differentiable almost everywhere in the state by Rademacher's
625 theorem, but a single trajectory is a null set and could in principle run inside
626 the exceptional set, so the chain rule step would be unavailable; recovering
627 the argument at that generality needs the Clarke generalised gradient in
628 place of ∇V . Smoothness is also, as it turns out, exactly what the natural
629 candidates lack.

630 We tried two. The value function of the Dinkelbach problem below sat-
631 isfies (25) with equality in the viscosity sense [25], but value functions of
632 this class are only Lipschitz. Computed on a 41^3 grid and tested on a 61^3
633 one, (25) fails at about 45% of test points by up to 2.9×10^{-3} . Halving λ
634 moves that violation by 0.6%, so the fault lies in the candidate and not in
635 the bound. At the opposite extreme an affine $V(x) = c_0 - a_0^\top x$ has an exactly
636 constant gradient and makes every requirement linear, so the best certifiable
637 λ is a linear program; over 167,936 constraints it returns $V \equiv 0$ and only
638 the pointwise bound $\min_{x,u} \ell$, which is vacuous here because the reachable
639 floor falls to within sampling resolution of zero at mid-life. Deleting the two
640 boundary requirements lets that program run to any cap imposed on it, which
641 identifies them as what binds: trajectories of (2) increase in every coordinate,
642 so an affine V is monotone along all of them at once and $V(x_0) \geq 0 \geq V|_{\text{failure}}$
643 forces it to decrease, while the near-aligned states, where the running cost
644 vanishes, demand a local increase.

645 The two failures are complementary, and together they say what a cer-
646 tificate needs. The value function varies its gradient exactly as the two de-
647 mands require but is not differentiable where that matters; affine functions
648 are perfectly smooth but cannot vary their gradient at all. A certificate must
649 therefore be both smooth and non-constant in its gradient, which points at
650 a smooth under-approximation of the value function, obtained by sum-of-
651 squares programming when the plant data are polynomial, or by interval
652 arithmetic with an explicit Lipschitz constant. We do not claim a certified
653 converse here, and the limit below remains a computation; what Proposi-
654 tion 5.1 buys is the replacement of an open problem without shape by a
655 concrete object with a checkable property.

656 Problems of the form (24) are handled by Dinkelbach’s transform [26]:
657 setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (26)$$

658 the map F is decreasing in λ and vanishes exactly at the optimal ratio, so λ^* is
659 found by bisection on $F(\lambda) = 0$. The inner problem in (26) has free terminal
660 time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman
661 equation on the state alone, with no time argument. Because $\dot{x} > 0$ the
662 state advances monotonically, and a semi-Lagrangian scheme [27] converges
663 by value iteration. One implementation detail matters: the scheme must use
664 a fixed spatial step rather than a fixed time step, because degradation speeds
665 vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at
666 one end of the envelope or the other.

667 6. Numerical study

668 Sections 3–5 are analytical, and three of their claims are stated sharply
669 enough to be tested rather than illustrated: that the loss is invisible to a
670 Fisher criterion, that the counting rule $m \geq d - 1$ binds in both direc-
671 tions on a generic plant, and that the best achievable loss is a finite number
672 one can compute. This section instantiates the model on a synthetic three-
673 mechanism plant on which every quantity is available in closed form, checks
674 those three claims against it, measures what steering is worth, and reads off
675 the dimensionless index that says in advance whether it will repay the effort.
676 Section 7 then reports what measured fleets can and cannot establish about
677 the premises.

6.1. The test plant

We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is the reciprocal temperature $1/(kT)$, so that this is the Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms. There is a single input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts zero loss out of reach unless this plant is one of the exceptions of Remark 4.5; Section 6.3 computes the floor and finds that it is not. Noise is unit and the signal-to-noise weight of Lemma 2.1 is $w = 0.9$. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of (3). The two values are equal by coincidence and play unrelated roles: w fixes the signal-to-noise ratio at which the loss is evaluated, x_f fixes where each trajectory stops. Raising x_f to unity leaves every qualitative conclusion below intact and moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° , and the gain of Section 6.5 falls from 68 to 58. No constant reported below should be read across thresholds.

Verification of the identities. Before using them we checked the analytical results numerically. The rotation identity (20) holds to relative error below 1.1×10^{-7} over the five input-channel pairs $(d, m) = (2, 1), (3, 2), (3, 1), (4, 2)$ and $(5, 3)$; the replicator form to 6.9×10^{-8} ; Theorem 3.6 to 5.1×10^{-6} , limited by finite differencing; and (17) to 1.1×10^{-16} .

6.2. A Fisher level set of varying loss

Proposition 4.6 is easy to demonstrate on this plant. For a constant input, the duty cycle only reparametrises time, so it moves $\|\mathbf{d}\|$ while leaving $\hat{\mathbf{d}}(\tau)$, and hence β , untouched: the Fisher score and the loss can be varied independently. Table 1 does exactly that, holding the Fisher score equal to seven digits by tuning the duty cycle δ . The loss still spans a factor 9.4.

The two objectives also disagree on which input to prefer: the raw score, before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and the loss is 1.7 times its value at $u = 1$. How far they disagree is a property of the instance; what is structural is only the blindness.

6.3. The reachability floor with a single input

With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} =$

Table 1: Loss of the fixed indicator at a Fisher score held equal. The duty cycle δ holds $\|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\ \mathbf{d}\ ^2$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	6.23	0.0215
1.0	0.628	0.4307565	4.53	0.0123
1.5	1.000	0.4307565	6.22	0.0208
2.0	1.564	0.4307565	7.53	0.0379
3.0	3.769	0.4307565	6.57	0.1161

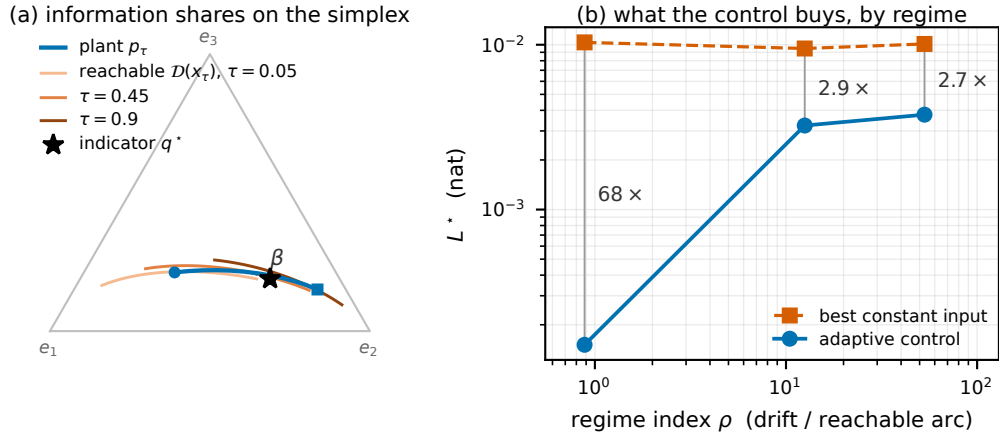


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* . No point lies on all three, so β cannot be held at zero (Theorem 4.1). (b) What steering buys, against the regime index ρ .

713 4.52°; it falls through mid-life to below the 0.01° resolution of our sampling,
714 where the indicator is momentarily reachable, and rises again towards failure
715 (Fig. 5(a)). Zero average loss would need the indicator reachable at every
716 instant and not at one, so it is unattainable however the input is steered, in
717 agreement with Theorem 4.1. Fig. 4(a) shows the geometry: the reachable
718 set sweeps across q^* as the plant ages, so the mid-life arc passes close to the
719 indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

720 6.4. The counting rule, tested in both directions

721 Theorem 4.1 is an equivalence, and the paragraph above tests only half
722 of it: with too few inputs the floor stays positive. The other half, that

supplying the inputs the counting rule asks for makes the floor vanish, is the more useful claim for a practitioner deciding how many operating variables to instrument, so we test it directly.

Give the same three-mechanism plant a second independent input by writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m = 2 = d - 1$ and each input acts on the mechanisms in a different pattern. Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction fixes the $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns, with a coefficient matrix of determinant 0.27 here, hence a unique solution at every state.

Taking as target the direction the plant occupies at mid-life under a central operating point, that solution lies inside the admissible box $[0, 3]^2$ at every one of 600 sampled states covering the whole life, and the residual misalignment there is 8.5×10^{-7} degrees, that is, zero to numerical precision; the schedule it requires is drawn in Fig. 5(b). An independent 90×90 grid search over the box returns a floor of at most 0.156° , which is the resolution of the grid rather than a property of the plant.

The contrast with the single-input case is the content of the theorem: the same mechanisms, the same shapes, the same envelope on each input, and the peak floor moves from 4.52° to zero at every sampled state on the addition of one operating variable. On this plant the counting rule is therefore not a conservative sufficient condition but a sharp one, and within the adopted model it fixes how many independent control degrees of freedom the plant must offer. How those degrees of freedom correspond to physical actuators, and how the mechanisms are individuated in the first place, are modelling questions this paper does not settle. Proposition 4.4 says which plants escape it, and Remark 4.5 says why a designer should not count on being one of them.

6.5. The value of steering

Minimising over q gives $q^* = (0.466, 0.771, 0.435)$, at which the adaptive policy attains 1.51×10^{-4} against 1.03×10^{-2} under the best constant input, a factor of 68. The ratio depends on how finely $\mathcal{U}$ is discretised for the policy, rising to 72 at 161 points; we report it on the 41-point grid used throughout.

The factor is an information ratio and is not free in reliability terms: the policy that attains it runs the component differently, and Section 6.6 finds the two closed loops separated by a factor of 3.4 in life. Aligning the indicator and prolonging the component are competing objectives, which is

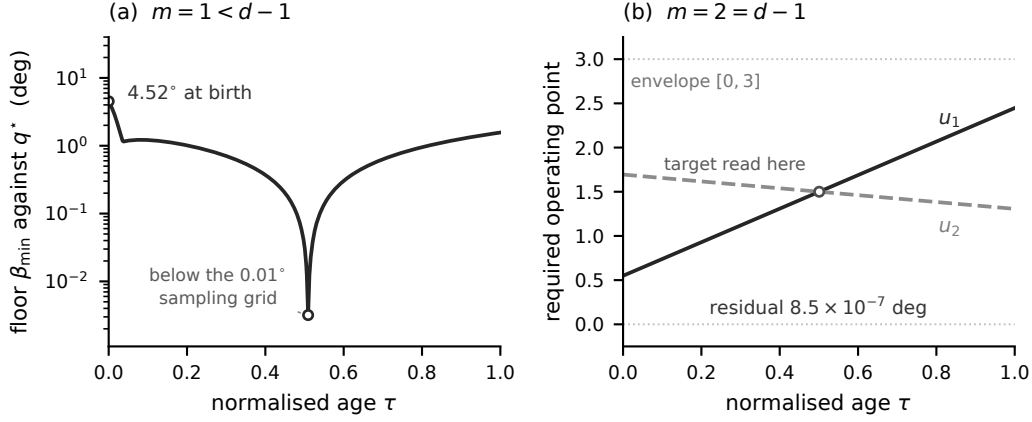


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly; plotted is the operating point this requires, which never leaves $\mathcal{U}$.

760 what makes the horizon in (24) a controlled quantity rather than a constant.
 761 That factor is the value of steering at all, not the value of steering optimally:
 762 a myopic policy that attends only to the numerator of (24) sits 2.7% above
 763 the optimum at q^* .

764 The fraction in (24) is doing visible work at this operating point. The
 765 closed loop at q^* and the closed loop at a balanced indicator, which calls for
 766 far less steering, differ by a factor 3.4 in lifetime: what alignment buys in the
 767 numerator is paid for in the denominator. This is the tension anticipated in
 768 Section 5, and it is why the horizon cannot be fixed in advance.

769 6.6. The fractional limit

770 Dinkelbach bisection with the control grid matched to the policy gives,
 771 over $N \in \{25, 29, 35\}$, extrapolants agreeing to 1.2% and $\lambda_\infty^* = 1.47 \times 10^{-4}$,
 772 below the realisable 1.51×10^{-4} as it must be.

773 We quote this to two digits only, for the following reason: the raw values
 774 are twice the extrapolant, so the first-order correction carries half the answer,
 775 and both quantities still move by 4% between control grids of 41 and 81
 776 points. Three checks guard the number: F must decrease and cross zero, λ^*
 777 must not exceed a realisable policy, and the grid sequence must extrapolate
 778 consistently. A fixed-horizon formulation failed the second and third and was
 779 discarded, as did a first attempt with mismatched control grids. A converse

Table 2: The regime index. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under adaptive control, and the gain its ratio to the best constant input.

E	arc	ρ	L^*	gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.5×10^{-4}	68×
(0.80, 0.82, 0.79)	1.93°	12.5	3.3×10^{-3}	2.9×
(0.80, 0.805, 0.798)	0.45°	53.1	3.8×10^{-3}	2.7×

bound uniform over trajectories, which would replace this computation by a theorem, remains open.

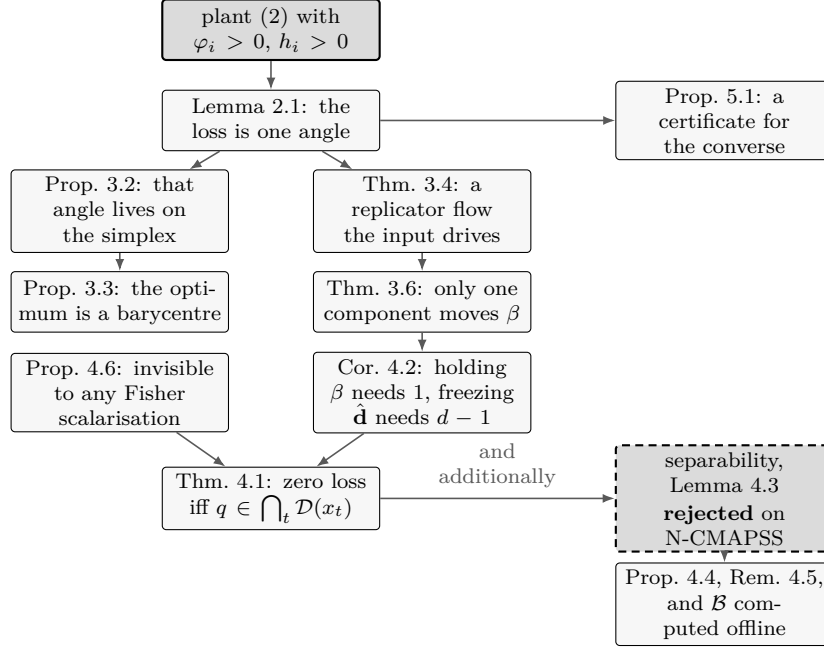
6.7. The regime index as a screening test

For a practitioner the first question is not how to steer but whether to bother. One dimensionless ratio answers it. The reachable set $\mathcal{D}(x)$ is an arc; write ρ for the drift of its midpoint over the life divided by its mean length, so ρ measures how far the informative direction wanders in units of the actuation authority available to chase it.

Table 2 varies the spread of the activation energies, which by (20) is what sets that authority. As the arc collapses, the attainable loss L^* rises by a factor 25 and the value of control falls from 68 to 2.7: with weak authority there is little left to steer. Fig. 4(b) plots the two against ρ . The construction of this paper is a statement about the low- ρ regime, and ρ can be computed from a nominal degradation model before any indicator is designed.

7. What public data can say about the premises

The construction rests on three modelling choices, and each is empirically testable. They do not carry equal weight, and Fig. 6 separates what depends on which. Two are premises about the plant: that the informative direction rotates along the trajectory, and that the input moves it. The third is the separability of Lemma 4.3, which is what makes the operator’s authority computable offline. This section summarises what six public run-to-failure fleets currently settle: the Severson lithium-ion cells [28], the XJTU-SY [29] and PRONOSTIA [30] bearing fleets, the NASA battery cells [31], and the C-MAPSS [32] and N-CMAPSS [33] turbofan simulations. The protocols are not reproduced here; the datasets are named so that each test can be located and repeated.



Every result on the spine needs only the positivity of the degradation model. Only the dashed branch needs separability, and it is that hypothesis, not the geometry, that Section 7 reports as rejected on the best public turbofan data.

Figure 6: What rests on what. The spine needs only the positivity of the degradation model. Only the dashed branch adds separability, which Section 7 reports as rejected on the best public turbofan data.

806 *Rotation.* On 129 Severson lithium-ion cells [28], with capacity fade, resis-
807 tance growth and mean temperature as the three mechanisms, the fleet-mean
808 informative direction travels 30.7° of arc over $\tau \in [0.15, 0.9]$ and ends 21°
809 from where it began (Fig. 2(c)). The direction stays in the open positive
810 orthant once the differentiating window reaches 301 cycles, so the positivity
811 that Remark 3.1 shows to be load-bearing is a property of this fleet and not
812 only of the model.

813 *Actuation.* This premise needs the operating point to vary while everything
814 else is held fixed, which of the six fleets only the two turbofan simulations
815 permit. On C-MAPSS [32], whose regimes interleave cycle by cycle inside
816 each engine so that an engine serves as its own control, the angle between
817 half-samples of different regimes exceeds the angle between half-samples of
818 the same regime by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439

819 engines. Against fake regimes cut from a single real one, where no contrast
820 exists by construction, the same machinery returns $|B - W| \leq 0.5^\circ$. On the
821 four measured fleets the obstacle is the estimator rather than the physics: in
822 the two bearing fleets the operating condition is fixed per unit, so “condition”
823 and “group of units” cannot be separated, and the NASA fleet has four cells.
824 The premise is therefore confirmed on simulation and open on hardware.

825 *Separability.* Lemma 4.3 is a statement about a plant class, and data can
826 refute it. If the shift between two operating regimes were the same at every
827 stage of life, regime and stage would act additively on ξ and a two-way layout
828 would show no interaction. On N-CMAPSS [33], which records altitude,
829 Mach number and throttle every second of real flight profiles, the interaction
830 exceeds *both* main effects on the sums of squares, by a factor 1.65 over the
831 operating-point effect on DS02 and 1.84 on DS01, and remains larger under
832 every clustering and staging tried. Separability is therefore rejected on the
833 highest-fidelity public turbofan simulation.

834 The consequence is the scope of the results above. The geometry of
835 Sections 3 and 4 needs only $\varphi_i > 0$ and $h_i > 0$, and is untouched. What
836 separability buys, and what the rejection withdraws, is Proposition 4.4 and
837 the offline route to $\mathcal{B}$: those describe a model class, and a plant must be
838 shown to belong to it before they are used. The two-way layout above is the
839 test, and it costs one pass over data a reliability programme already holds.

840 8. Discussion

841 Four practical readings follow from the results above. The first two hold
842 whatever the plant is, because one is a test and the other an invariance. The
843 last two assume the separable form of (2), which Section 7 reports as rejected
844 on the only data able to test it, so they apply where that form does and not
845 otherwise. The distinction matters more than any of the four.

846 Test the separability assumption before relying on it. The two-way layout
847 of Section 7 needs only a fleet whose operating point varies within a unit, and
848 it returns an interaction against a measured noise floor. If the interaction
849 clears that floor, the stress model is not separable on that fleet, the offline
850 route to the operator’s authority is unavailable, and the two model-based
851 readings below do not apply. The test costs one pass over data the programme
852 already holds.

853 A Fisher criterion cannot be used to design a fixed indicator. This is the
854 second of the two unconditional readings: Proposition 4.6 is an invariance of

855 the loss under a common gain, and holds whatever the plant. Design rules
 856 that maximise a scalarisation of the Fisher information, a natural and com-
 857 mon choice, are provably unable to distinguish configurations whose losses
 858 differ by an order of magnitude, because they measure the strength of the
 859 signal and not the direction it points in. If the two objectives coincide in a
 860 given application, that is a property of the application and should be verified,
 861 not assumed.

862 The binding constraint is actuation authority, not indicator sophistica-
 863 tion. The counting rule of Theorem 4.1 prices the operating variables a
 864 lossless fixed indicator would need, and Proposition 4.4 sharpens it: the
 865 count is sufficient and not necessary, but the exceptions ask the plant’s log-
 866 offsets to slide along the input’s own direction, which Remark 4.5 shows to
 867 be a knife edge. Care in choosing the weights substitutes for actuation in
 868 neither case. What one controllable variable does buy is the weaker objective
 869 of Corollary 4.2: an installation that could never freeze the informative direc-
 870 tion can still hold the misalignment constant, though not at zero. Authority
 871 bought this way is not free either. On the test plant the indicator whose loss
 872 is smallest is served by a policy that changes the component’s life by a factor
 873 of 3.4 against a balanced indicator, so information-preserving operation and
 874 long life are competing objectives and the trade between them is a reliability
 875 decision rather than an estimation one.

876 The decision to steer precedes the design of the indicator. The regime
 877 index ρ of Section 6.7 requires only a nominal degradation model and the
 878 operating envelope. When ρ is large, the reachable arc is short relative to
 879 the drift, and Table 2 shows the value of steering collapsing from $68\times$ to
 880 under $3\times$. The index is therefore a screening test to run before committing
 881 engineering effort.

882 *Limitations.* Three limitations qualify the readings above. The separable
 883 form of (2) is not only an idealisation but one the data reject: Section 7 finds
 884 the interaction between operating point and state larger than either main
 885 effect on two N-CMAPSS datasets, so computing the operator’s authority
 886 offline is available only where that form holds. The fundamental limit of
 887 Section 5 is computed rather than proved, to two digits. And the actuation
 888 premise is confirmed on simulated rather than measured data, for the reasons
 889 set out in Section 7.

890 9. Conclusion

891 A fixed health indicator discards information, and this paper has treated
892 what it discards as a quantity the operating point controls. Indicator design
893 then becomes a reachability question. The family of reachable direction sets
894 bounds what the operating point can achieve, and their drift relative to their
895 width is a dimensionless measure of whether substantial benefit is plausi-
896 ble. That ratio is dimensionless and available before the designer fixes any
897 indicator. Actuation authority, rather than sophistication of the indicator,
898 governs whether a fixed compression can come close to lossless. Whether
899 that authority is present is an empirical question rather than a modelling
900 assumption, and on 439 simulated turbofan records it is. Establishing the
901 same on measured fleets awaits records in which the operating point varies
902 enough to show the effect.

903 The separable plant form that makes all of this computable is rejected by
904 the data, on two datasets of the highest-fidelity public turbofan simulation,
905 so the geometry of Section 4.1 describes a model class rather than measured
906 machinery, and the offline route it opens is available only where that class
907 applies. What replaces it when the stress model is not separable, and whether
908 a weaker structure still yields a computable reachability criterion, is the
909 question this leaves.

910 Two further questions remain open, the first of them now in a more
911 specific form. Proposition 5.1 reduces the search for a converse bound uni-
912 form over trajectories to the exhibition of a single continuously differentiable
913 verification function; the value function is the natural candidate and fails
914 the differential test for a structural reason, so what is required is a smooth
915 under-approximation of it, by sum-of-squares programming or by interval
916 arithmetic with an explicit Lipschitz constant. The other is unchanged: the
917 fractional limit to better than two digits, which requires a scheme of higher
918 order than the semi-Lagrangian one used here.

919 Data availability and reproducibility

920 The six fleets named in Section 7 are publicly available. One script re-
921 generates every computed result from a single configuration, and a compan-
922 ion check recomputes each reported number and compares it with the value
923 written in the text, covering the tables and the quantities reported alongside
924 them. Those checks are maintained against the full manuscript from which

925 this paper is drawn, and all of them pass; the numbers reported here are
926 carried over from it unchanged. Three groups of numbers lie outside it: the
927 residuals of the identity checks in Section 6, the barycentre comparison in
928 Section 3.3, and the diagnostics of the two discarded formulations in Sec-
929 tion 5. The plots are produced by three further scripts. All of these are
930 available from the authors.

931 **Declaration of competing interest**

932 The authors declare that they have no known competing financial inter-
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